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Cascaded Pipeline: VAD → ASR → LLM → TTS

Data Flow

1. **Voice Activity Detection (VAD)**: Detects speech onset/offset to segment audio into utterances. Modern systems use **semantic VAD** that waits for linguistically complete turns rather than silence gaps. [inworld](#)
2. **ASR (Automatic Speech Recognition)**: Streams partial transcripts while user speaks, finalizes on turn end. Outputs: text transcript + confidence scores. [livekit](#)
3. **LLM**: Receives transcript, generates text response token-by-token. Can invoke tools, access RAG, apply guardrails mid-generation. [inworld](#)
4. **TTS (Text-to-Speech)**: Converts LLM tokens to audio, streams first audio chunk before LLM finishes. [deepgram](#)

```
User audio → VAD → ASR (text) → LLM (text) → TTS → Agent audio
                ↑           ↑           ↑
            transcript  tool calls  voice synthesis
```

Where Latency Accumulates

STAGE	TYPICAL LATENCY (P50)	NOTES
VAD end-of-speech detection	20–100 ms	Semantic VAD reduces false endpoints gradium
ASR first partial	100–300 ms	Streaming; final transcript adds 50–100 ms livekit
LLM time-to-first-token (TTFT)	200–600 ms	Dominates pipeline; depends on model size deepgram
TTS time-to-first-audio (TTFA)	75–300 ms	Modern APIs (e.g., Realtime TTS-2 Flash) achieve <100 ms P99 inworld
Network + orchestration	50–200 ms	WebRTC <50 ms; PSTN 150–700 ms livekit
Total TTFA	600–1,200 ms	Sub-500 ms achievable with aggressive streaming futureagi

Latency is **additive**: each stage must wait for prior stage output, though streaming overlaps ASR→LLM→TTS partially. [livekit](#)

Information Lost at ASR Step

ASR discards **all paralinguistic information**:

- **Tone**: pitch contour, intonation patterns (question vs. statement)
- **Emotion**: frustration, excitement, sarcasm, uncertainty
- **Emphasis**: which words are stressed for contrast or focus
- **Hesitation**: filled pauses ("um", "uh"), elongated vowels, restarts
- **Speaker identity**: voice timbre, accent, age/gender cues
- **Background context**: ambient noise, overlapping speech, laughter [inworld](#)

Example: User says "***I'm fine***" with flat tone and elongated "fine" (indicating frustration). ASR outputs "I'm fine" —the LLM cannot distinguish genuine vs. ironic fine. [gradium](#)

What Cascaded Pipelines Can Do With Paralinguistic Cues

Nothing directly. The LLM reasons over text only. Some systems attempt reconstruction:

- TTS can add prosody based on punctuation or SSML tags, but this is synthetic, not responsive to user emotion. [inworld](#)
- Separate emotion classifiers on audio (pre-ASR) can inject metadata, but this is ad-hoc and rarely production-grade. [gradium](#)

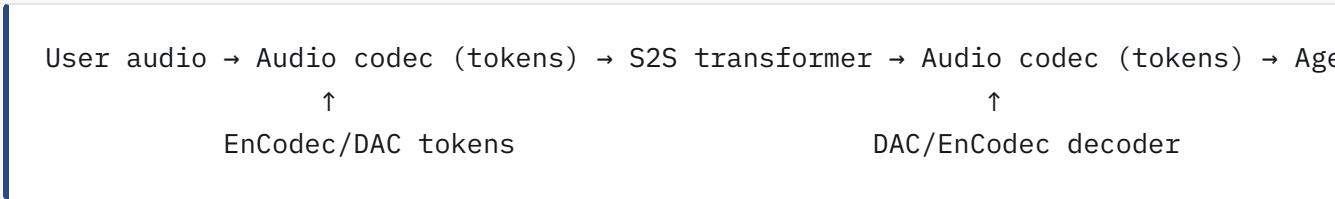
Controllability, Debuggability, Cost, Tool Calls, Guardrails

DIMENSION	CASCADED PIPELINE
Controllability	High: swap STT/LLM/TTS independently; tune each component; apply SSML to TTS; prompt-engineer LLM inworld
Debuggability	High: per-stage spans (ASR WER, LLM hallucination, TTS mispronunciation); transcript-native logging for compliance inworld
Cost	Per-component pricing: STT ~\$0.15/hr, TTS ~\$25/1M chars, LLM at-cost routing. Can downgrade LLM without touching voice quality. inworld
Tool calls	Mature: inherits LLM provider's full function-calling surface (OpenAI, Anthropic). Supports multi-tool chains, strict JSON schema, parallel calls. inworld
Guardrails mid-turn	Easy: intercept transcript pre-LLM (PII redaction, moderation); intercept LLM tokens pre-TTS (safety filters, RAG injection). inworld

End-to-End Speech-to-Speech Models

Data Flow

Single multimodal model consumes audio tokens, emits audio tokens directly:



No intermediate transcript on hot path. Transcript may be generated for logging via separate ASR pass. [inworld](#)

Key systems:

- **OpenAI Realtime API** (`gpt-realtime`): WebRTC/WebSocket, function calling, multi-modal (audio + text + image). [futureagi](#)
- **Google Gemini Live API**: Native audio/video/text input, tool calling, long-context conversation. [futureagi](#)
- **Moshi** (Kyutai, 2024): Open-weight, full-duplex (two simultaneous audio streams), no turn-taking mechanism. [gradium](#)

Where Latency Accumulates

COMPONENT	TYPICAL LATENCY	NOTES
Single inference pass (audio-in → audio-out)	300–500 ms TTFA	No STT/LLM/TTS handoff; warm WebRTC connection inworld
Network + transport	50–150 ms	WebRTC optimized livekit
Total TTFA	300–500 ms	~2× faster than typical cascaded inworld

Latency is **non-additive**: single forward pass avoids sequential bottlenecks. However, large multimodal models may have higher per-token compute cost. [inworld](#)

Paralinguistic Information Preserved

S2S models perceive and respond to:

- **Tone**: matches output prosody to user's pitch contour (e.g., gentle response to flat tone). [inworld](#)
- **Emotion**: detects frustration, excitement, sarcasm from audio features; modulates response accordingly. [inworld](#)
- **Emphasis**: hears which words user stressed; can mirror or contrast in response. [inworld](#)
- **Hesitation**: detects uncertainty from filled pauses, elongations; can offer clarification or encouragement. [inworld](#)

- **Speaker identity:** recognizes voice timbre across turns; enables personalized voice cloning without explicit enrollment. [inworld](#)
- **Background context:** laughter, overlapping speech, ambient noise inform turn-taking and response timing. [inworld](#)

Example: User says **"*I'm fine*"** with frustrated tone. S2S model hears frustration, responds gently: **"*You sound upset—want to talk about it?*"** [inworld](#)

Full-Duplex Conversation

Moshi and similar models process **two simultaneous audio streams** (user + agent), eliminating turn-taking:

- No VAD required; model can respond while user speaks (backchanneling: "mhm", "uh-huh"). [gradium](#)
- Natural overlap handling: user interrupts, model pauses, both speak briefly. [gradium](#)

Controllability, Debuggability, Cost, Tool Calls, Guardrails

DIMENSION	END-TO-END S2S
Controllability	Low: LLM baked into weights; cannot swap components; fixed voice sets; prompt engineering not directly applicable. inworld
Debuggability	Low: single audio-in/audio-out span; no per-stage transcript; must sample-decode to text for retrospective eval. inworld
Cost	Bundled audio-token pricing: OpenAI gpt-realtime-2.1 at \$32/1M input tokens + \$64/1M output tokens; Gemini Live at \$0.005/min in + \$0.018/min out. No per-component levers. inworld
Tool calls	Emerging: OpenAI Realtime and Gemini Live support function calling, but edge cases (mid-turn barge-in during tool call, strict schema validation) are rougher than cascaded. inworld
Guardrails mid-turn	Limited: no transcript boundary to intercept; must run inline audio moderation (e.g., Future AGI Protect on output audio). futureagi

Trade-Off Summary

AXIS	CASCADED WINS WHEN...	S2S WINS WHEN...
Latency	Aggressive streaming + co-located components achieve sub-500 ms futureagi	Default 300–500 ms without tuning; critical for conversational UX inworld
Paralinguistic awareness	Not required (transactional: scheduling, support, healthcare) inworld	Core product (companions, language tutors, social apps) inworld
Tool depth	5+ tools with strict schemas (booking, payments, PHI) inworld	1–3 tools with loose validation (search, lookup) futureagi
Observability	Compliance, moderation, eval require transcript-native logging inworld	Prototype/research; can tolerate post-hoc sampled transcript eval futureagi
Vendor flexibility	Need to swap LLM quarterly; multi-vendor STT/TTS inworld	Single-vendor simplicity acceptable; locked to S2S provider inworld
Voice cloning	Custom brand voice required (ElevenLabs, Cartesia) inworld	Fixed voice sets sufficient (OpenAI shimmer/alloy/echo) inworld

Key Papers and System Cards

- **Moshi** (Kyutai, 2024, arXiv:2410.00037): First open full-duplex S2S model with two simultaneous audio streams. [gradium](#)
- **OpenAI Realtime API** (2024–2026): Production S2S with function calling, multi-modal input. Pricing: \$32/1M input tokens, \$64/1M output tokens (gpt-realtime-2.1). [inworld](#)
- **Gemini Live API** (Google, 2025–2026): Native audio/video/text S2S with tool calling. Pricing: \$0.005/min in, \$0.018/min out. [inworld](#)
- **SoundStream** (Zeghidour et al., 2021, arXiv:2107.03312): Neural audio codec enabling discrete audio tokens for S2S. [gradium](#)
- **Cascaded vs. S2S comparisons**: Inworld AI (July 2026), Gradium (Sept 2026), Future AGI (April 2026). [inworld](#)

The field is converging toward **hybrid architectures**: semantic VAD + streaming cascaded pipelines with S2S-like prosody transfer, or S2S models with modular LLM backends. Until

then, cascaded remains the production default; S2S leads in conversational naturalness.
[inworld](#)